



STUDENT PERFORMANCE PREDICTION SYSTEM USING MACHINE LEARNING

Technologies Used- Python, Scikit-learn, Pandas, NumPy, Streamlit

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PROBLEM STATEMENT



Educational institutions generate large amounts of student data, but it is often underutilised. Predicting student performance can help identify strengths and weaknesses early, allowing students to take corrective actions and improve academic outcomes.

PROJECT OBJECTIVES

- Predict student academic performance using Machine Learning.
- Analyze factors affecting student success.
- Compare multiple classification algorithms.
- Identify important features influencing performance.
- Develop an interactive web-based dashboard.
- Provide personalized recommendations to students.





DATASET AND FEATURES

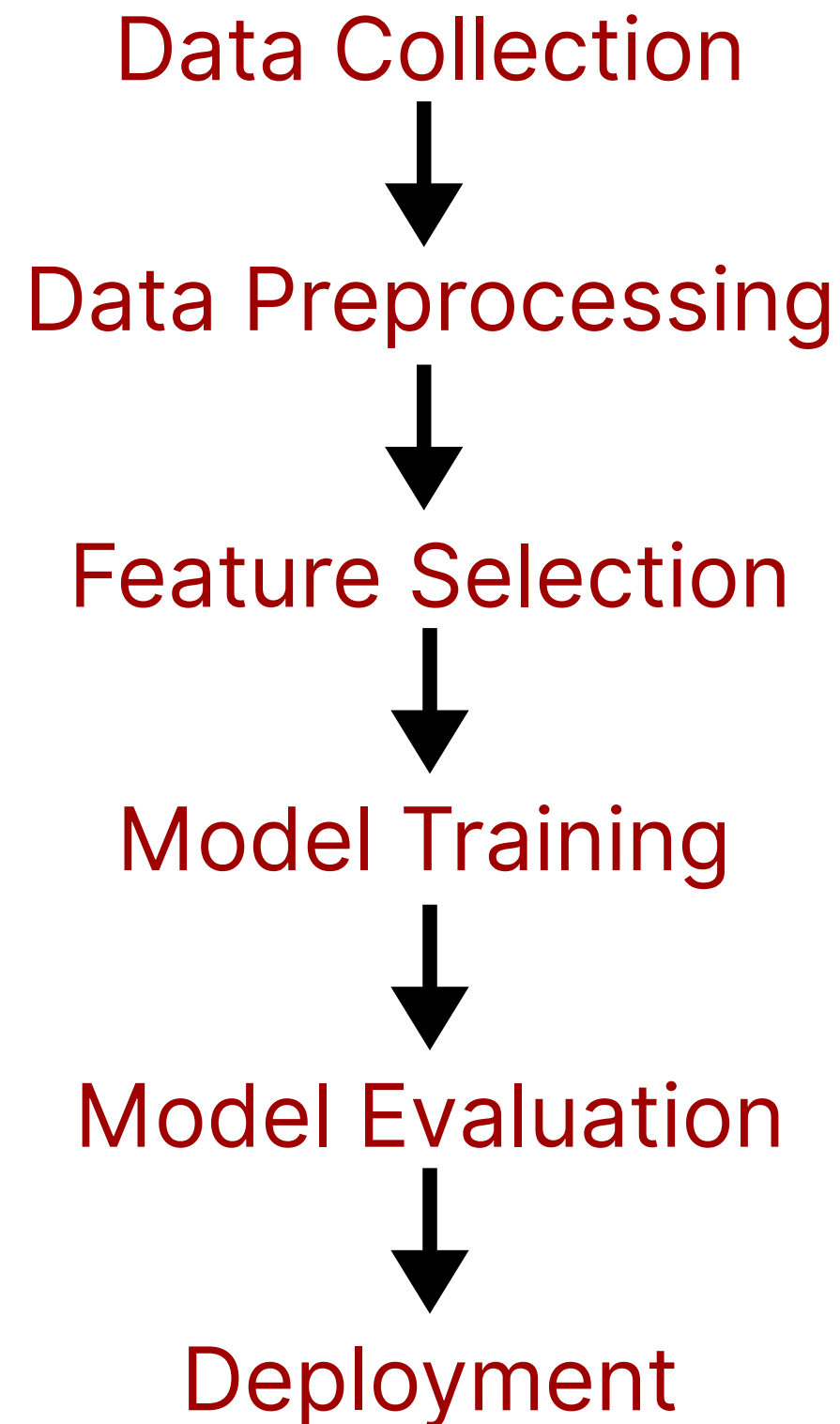
Academic and behavioral attributes were used:

- Study Hours
- Attendance
- Learning Resources
- Extracurricular Activities
- Motivation Level
- Internet Access
- Online Courses
- Discussion Participation
- Assignment Completion
- Educational Technology Usage
- Stress Level

Student Performance Category (Target Variable):

- Excellent
- Good
- Average
- Poor

PROJECT WORKFLOW





FEATURE SELECTION AND MODEL TRAINING

FEATURE SELECTION

Initially, all available features were used. Feature importance analysis was performed to identify the most influential variables and reduce unnecessary complexity.

MODEL EVALUATED

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

Multiple iterations were performed to improve prediction quality and model interpretability.

MODEL COMPARISON

Model	Accuracy
Logistic Regression	50.08%
Decision Tree	89.01%
Random Forest	86.08%

Hence, the Selected Model is **Decision Tree Classifier**

Reasons:

- Highest Accuracy
- Better Interpretability
- Suitable for Deployment

KEY INSIGHTS

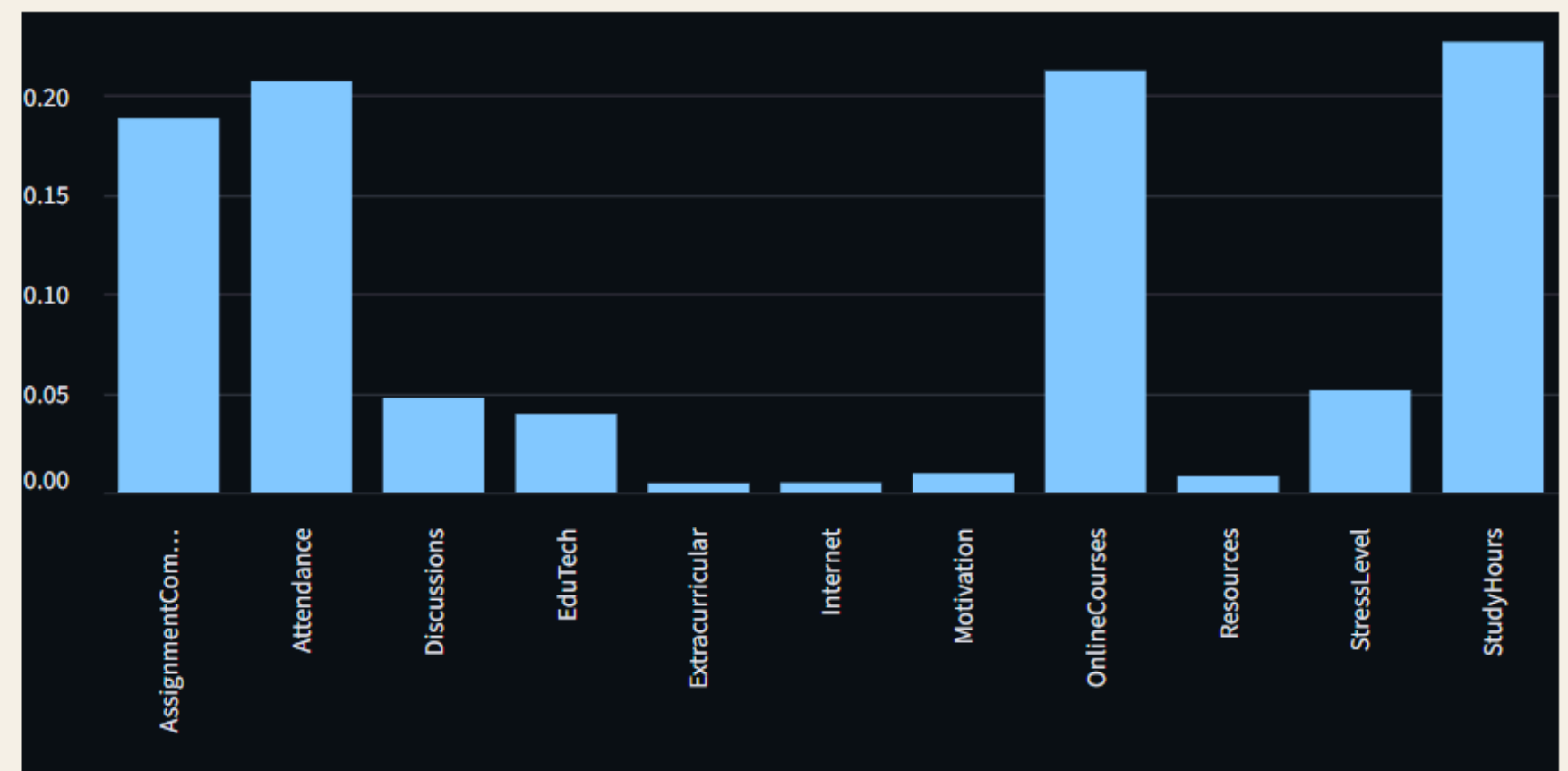
Important Predictors:

- Assignment Completion
- Attendance
- Study Hours
- Online Courses

Observations:

Students with better attendance, assignment completion, and study habits generally achieved higher performance levels

 Feature Importance



DASHBOARD AND DEPLOYMENT

Streamlit Dashboard Features:

- User-friendly interface
- Real-time performance prediction
- Personalized recommendations
- Feature importance visualization
- Interactive input controls

Deployment

- GitHub Repository
- Streamlit Cloud Deployment

The screenshot shows a web application titled "Student Performance Prediction System" with a subtitle "AI-powered academic performance prediction using Machine Learning techniques." The dashboard is divided into two columns of input controls. The left column includes sliders for "Study Hours" (set to 4), "Attendance (%)" (set to 75), "Learning Resources" (set to 5), "Extracurricular Participation" (set to "No" via a dropdown), and "Motivation Level" (set to 5). The right column includes dropdowns for "Internet Access" (set to "No") and "Use of Educational Technology" (set to "No"), and sliders for "Online Courses Completed" (set to 5), "Discussion Participation" (set to 5), "Assignment Completion (%)" (set to 70), and "Stress Level" (set to 5). At the bottom, there is a "Predict Performance" button and a "Main" button on the right.

Feature	Value
Study Hours	4
Attendance (%)	75
Learning Resources	5
Extracurricular Participation	No
Motivation Level	5
Internet Access	No
Online Courses Completed	5
Discussion Participation	5
Assignment Completion (%)	70
Use of Educational Technology	No
Stress Level	5

RESULTS, CONCLUSION & FUTURE SCOPE



Results

- Decision Tree achieved 89.01% accuracy.
- Feature selection improved model quality.
- Real-time prediction system successfully deployed.

Future Scope

- Integration with real educational databases
- Enhanced recommendation systems
- Advanced analytics and reporting
- Institution-level implementation

Conclusion

A Machine Learning-based Student Performance Prediction System was successfully developed and deployed. The system provides academic performance predictions and recommendations through an interactive dashboard.



THANK YOU